MZUZU UNIVERSITY

DEPARTMENT OF MATHEMATICS

EMT 1201-Calculus

Supplementary Examination

Date: 12th June, 2017

Time Allowed: 3 hours (8:00 - 11:00)

Instructions:

- Show all your working.
- Attempt all questions.

1. (a) Given that $\lim_{x\to 2} f(x) = 4$ and $\lim_{x\to 2} g(x) = -2$. Find the limit $\lim_{x\to 2} \frac{3f(x)}{g(x)}$

[2 marks]

(b) Evaluate the limit $\lim_{u\to -2} \sqrt{u^4 + 3u + 6}$

[2 marks]

(c) If $\lim_{x\to 1} \dfrac{f(x) - 8}{x - 1} = 10$, find $\lim_{x\to 1} f(x)$

[3 marks]

2. (a) Let $g(x) = \dfrac{x^2 + x - 6}{|x - 2|}$. Show that the limit $\lim_{x\to 2} g(x)$ does not exist.

[6 marks]

(b) If $4x - 9 \leq f(x) \leq x^2 - 4x + 7$ for $x \geq 0$, find $\lim_{x\to 4} f(x)$

[3 marks]

3. (a) Find the derivative of $g(x) = \sqrt{1 + 2x}$, by evaluating the limit

$$\lim_{h\to 0} \frac{f(x + h) - f(x)}{h}$$

[4 marks]

(b) Differentiate:

i. $f(x) = x^8 + 12x^5 - 4x^4 + 10x^3 - 6x + 5$

[2 marks]

ii. $f(x) = \dfrac{x^3}{1 - x^2}$

[4 marks]

iii. $f(x) = \sqrt{1 + 2e^{3x}}$.

[4 marks]

(c) If $f(x) = \sqrt{x}g(x)$, where $g(4) = 2$ and $g'(4) = 3$, Find $f'(4)$

[4 marks]

4. (a) Find the horizontal and vertical asymptotes of $\dfrac{2x + 1}{x - 2}$.

[6 marks]

(b) compute $\lim_{x\to\infty} (\sqrt{x^2 + 1} - x)$

[5 marks]

5. (a) Use the definition of continuity and the properties of limits to show that the function $f(x) = (x + 2x^3)^4$ is continuous at -1.

[4 marks]

(b) Suppose f and g are continuous functions such that $g(2) = 6$ and

$$\lim_{x \to 2}[3f(x) + f(x)g(x)] = 36.$$

Find $f(2)$.

[3 marks]

6. Given that $x^3 + y^3 = 6xy$. Find;

(a) y'

[3 marks]

(b) the tangent to the folium of Descartes $x^3 + y^3 = 6xy$ at the point $(3, 3)$

[3 marks]

7. (a) Find the derivative of $y = \ln(e^{-x} + xe^{-x})$ with respect to x.

[4 marks]

(b) Use logarithmic differentiation to find the derivative of the function

$$y = (2x + 1)^5(x^4 - 3)^6$$

[6 marks]

8. (a) Find the critical numbers of the function $f(x) = x^{\frac{4}{5}}(x - 2)^2$

[6 marks]

(b) Find the absolute maximum and minimum values of $f(x) = 2x^3 - 3x^2 - 12x + 1$ on the interval $[-2, 3]$.

[6 marks]

END OF QUESTIONS
GOOD LUCK!!